UNIT-3

Process Synchronization in OS (Operating System)

When two or more process cooperates with each other, their order of execution must be preserved otherwise there can be conflicts in their execution and inappropriate outputs can be produced.

A cooperative process is the one which can affect the execution of other process or can be affected by the execution of other process. Such processes need to be synchronized so that their order of execution can be guaranteed.

The procedure involved in preserving the appropriate order of execution of cooperative processes is known as Process Synchronization. There are various synchronization mechanisms that are used to synchronize the processes.

Race Condition

A Race Condition typically occurs when two or more threads try to read, write and possibly make the decisions based on the memory that they are accessing concurrently.

Critical Section

The regions of a program that try to access shared resources and may cause race conditions are called critical section. To avoid race condition among the processes, we need to assure that only one process at a time can execute within the critical section.

Critical Section Problem in OS (Operating System)

Critical Section is the part of a program which tries to access shared resources. That resource may be any resource in a computer like a memory location, Data structure, CPU or any IO device.

The critical section cannot be executed by more than one process at the same time; operating system faces the difficulties in allowing and disallowing the processes from entering the critical section.

The critical section problem is used to design a set of protocols which can ensure that the Race condition among the processes will never arise.

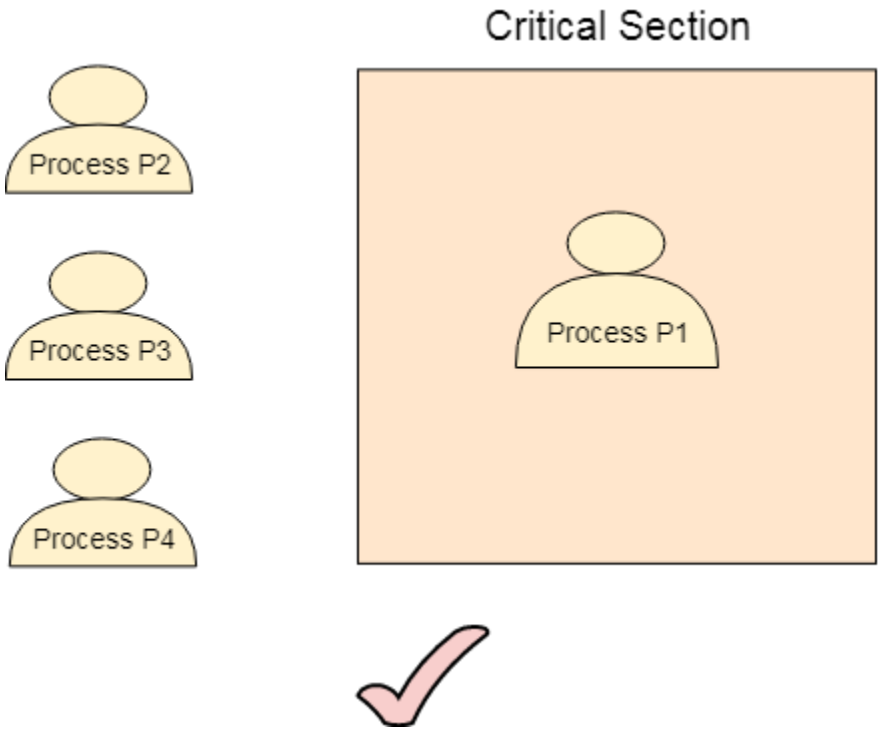
In order to synchronize the cooperative processes, our main task is to solve the critical section problem. We need to provide a solution in such a way that the following conditions can be satisfied.

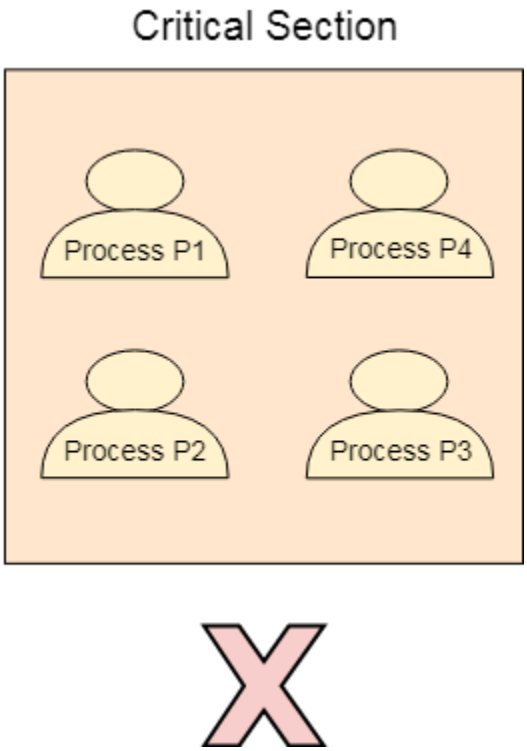
Requirements of Synchronization mechanisms

Primary

1. **Mutual Exclusion**

   Our solution must provide mutual exclusion. By Mutual Exclusion, we mean that if one process is executing inside critical section then the other process must not enter in the critical section.





2. **Progress**

   Progress means that if one process doesn't need to execute into critical section then it should not stop other processes to get into the critical section.

Secondary

1. **Bounded Waiting**

   We should be able to predict the waiting time for every process to get into the critical section. The process must not be endlessly waiting for getting into the critical section.

2. **Architectural Neutrality**

   Our mechanism must be architectural natural. It means that if our solution is working fine on one architecture then it should also run on the other ones as well.

Lock Variable

This is the simplest synchronization mechanism. This is a Software Mechanism implemented in User mode. This is a busy waiting solution which can be used for more than two processes.

In this mechanism, a Lock variable **lock**is used. Two values of lock can be possible, either 0 or 1. Lock value 0 means that the critical section is vacant while the lock value 1 means that it is occupied.

A process which wants to get into the critical section first checks the value of the lock variable. If it is 0 then it sets the value of lock as 1 and enters into the critical section, otherwise it waits.

The pseudo code of the mechanism looks like following.

```
1. Entry Section →
2. While (lock! = 0);
3. Lock = 1;
4. //Critical Section
5. Exit Section →
6. Lock =0;
```

If we look at the Pseudo Code, we find that there are three sections in the code. Entry Section, Critical Section and the exit section.

Initially the value of **lock variable** is **0**. The process which needs to get into the **critical section**, enters into the entry section and checks the condition provided in the while loop.

The process will wait infinitely until the value of **lock** is 1 (that is implied by while loop). Since, at the very first time critical section is vacant hence the process will enter the critical section by setting the lock variable as 1.

When the process exits from the critical section, then in the exit section, it reassigns the value of **lock** as 0.

Every Synchronization mechanism is judged on the basis of four conditions.

1. Mutual Exclusion
2. Progress
3. Bounded Waiting
4. Portability

Out of the four parameters, Mutual Exclusion and Progress must be provided by any solution. Let?s analyze this mechanism on the basis of the above mentioned conditions.

Mutual Exclusion

The lock variable mechanism doesn't provide Mutual Exclusion in some of the cases. This can be better described by looking at the pseudo code by the Operating System point of view I.E. Assembly code of the program. Let's convert the Code into the assembly language.

1. *Load Lock, R0*
2. *CMP R0, #0*
3. *JNZ Step 1*
4. *Store #1, Lock*
5. *Store #0, Lock*

Let us consider that we have two processes P1 and P2. The process P1 wants to execute its critical section. P1 gets into the entry section. Since the value of lock is 0 hence P1 changes its value from 0 to 1 and enters into the critical section.

Meanwhile, P1 is preempted by the CPU and P2 gets scheduled. Now there is no other process in the critical section and the value of lock variable is 0. P2 also wants to execute its critical section. It enters into the critical section by setting the lock variable to 1.

Now, CPU changes P1's state from waiting to running. P1 is yet to finish its critical section. P1 has already checked the value of lock variable and remembers that its value was 0 when it previously checked it. Hence, it also enters into the critical section without checking the updated value of lock variable.

Now, we got two processes in the critical section. According to the condition of mutual exclusion, morethan one process in the critical section must not be present at the same time. Hence, the lock variable mechanism doesn't guarantee the mutual exclusion.

The problem with the lock variable mechanism is that, at the same time, more than one process can see the vacant tag and more than one process can enter in the critical section. Hence, the lock variable doesn't provide the mutual exclusion that's why it cannot be used in general.

Since, this method is failed at the basic step; hence, there is no need to talk about the other conditions to be fulfilled.

Test Set Lock Mechanism

Modification in the assembly code

In lock variable mechanism, Sometimes Process reads the old value of lock variable and enters the critical section. Due to this reason, more than one process might get into critical section. However, the code shown in the part one of the following section can be replaced with the code shown in the part two. This doesn't affect the algorithm but, by doing this, we can manage to provide the mutual exclusion to some extent but not completely.

In the updated version of code, the value of Lock is loaded into the local register R0 and then value of lock is set to 1.

However, in step 3, the previous value of lock (that is now stored into R0) is compared with 0. if this is 0 then the process will simply enter into the critical section otherwise will wait by executing continuously in the loop.

The benefit of setting the lock immediately to 1 by the process itself is that, now the process which enters into the critical section carries the updated value of lock variable that is 1.

In the case when it gets preempted and scheduled again then also it will not enter the critical section regardless of the current value of the lock variable as it already knows what the updated value of lock variable is.

Section 1	**Section 2**
1. Load Lock, R0 2. CMP R0, #0 3. JNZ step1 4. store #1, Lock	1. Load Lock, R0 2. Store #1, Lock 3. CMP R0, #0 4. JNZ step 1

TSL Instruction

However, the solution provided in the above segment provides mutual exclusion to some extent but it doesn't make sure that the mutual exclusion will always be there. There is a possibility of having more than one process in the critical section.

What if the process gets preempted just after executing the first instruction of the assembly code written in section 2? In that case, it will carry the old value of lock

variable with it and it will enter into the critical section regardless of knowing the current value of lock variable. This may make the two processes present in the critical section at the same time.

To get rid of this problem, we have to make sure that the preemption must not take place just after loading the previous value of lock variable and before setting it to 1. The problem can be solved if we can be able to merge the first two instructions.

In order to address the problem, the operating system provides a special instruction called **Test Set Lock (TSL)** instruction which simply loads the value of lock variable into the local register R0 and sets it to 1 simultaneously

The process which executes the TSL first will enter into the critical section and no other process after that can enter until the first process comes out. No process can execute the critical section even in the case of preemption of the first process.

The assembly code of the solution will look like following.

1. *TSL Lock, R0*
2. *CMP R0, #0*
3. *JNZ step 1*

Let's examine TSL on the basis of the four conditions.

- **Mutual Exclusion**

  Mutual Exclusion is guaranteed in TSL mechanism since a process can never be preempted just before setting the lock variable. Only one process can see the lock variable as 0 at a particular time and that's why, the mutual exclusion is guaranteed.

- **Progress**

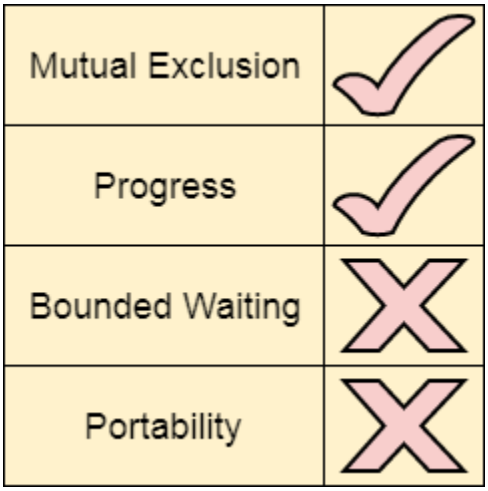
  According to the definition of the progress, a process which doesn't want to enter in the critical section should not stop other processes to get into it. In TSL mechanism, a process will execute the TSL instruction only when it wants to get into the critical section. The value of the lock will always be 0 if no process doesn't want to enter into the critical section hence the progress is always guaranteed in TSL.

- **Bounded Waiting**

  Bounded Waiting is not guaranteed in TSL. Some process might not get a chance for so long. We cannot predict for a process that it will definitely get a chance to enter in critical section after a certain time.

- **Architectural Neutrality**

  TSL doesn't provide Architectural Neutrality. It depends on the hardware platform. The TSL instruction is provided by the operating system. Some platforms might not provide that. Hence it is not Architectural natural.



Priority Inversion

In TSL mechanism, there can be a problem of priority inversion. Let?s say that there are two cooperative processes, P1 and P2.

The priority of P1 is 2 while that of P2 is 1. P1 arrives earlier and got scheduled by the CPU. Since it is a cooperative process and wants to execute in the critical section hence it will enter in the critical section by setting the lock variable to 1.

Now, P2 arrives in the ready queue. The priority of P2 is higher than P1 hence according to priority scheduling, P2 is scheduled and P1 got preempted. P2 is also a cooperative process and wants to execute inside the critical section.

Although, P1 got preempted but it the value of lock variable will be shown as 1 since P1 is not completed and it is yet to finish its critical section.

P1 needs to finish the critical section but according to the scheduling algorithm, CPU is with P2. P2 wants to execute in the critical section, but according to the synchronization mechanism, critical section is with P1.

This is a kind of lock where each of the process neither executes nor completes. Such kind of lock is called **Spin Lock**.

This is different from deadlock since they are not in blocked state. One is in ready state and the other is in running state, but neither of the two is being executed.

Turn Variable or Strict Alternation Approach

Turn Variable or Strict Alternation Approach is the software mechanism implemented at user mode. It is a busy waiting solution which can be implemented only for two processes. In this approach, A turn variable is used which is actually a lock.

This approach can only be used for only two processes. In general, let the two processes be Pi and Pj. They share a variable called turn variable. The pseudo code of the program can be given as following.

For Process Pi

1. Non - CS
2. while (turn ! = i);
3. Critical Section
4. turn = j;
5. Non - CS

For Process Pj

1. Non - CS
2. while (turn ! = j);
3. Critical Section
4. turn = i ;
5. Non - CS

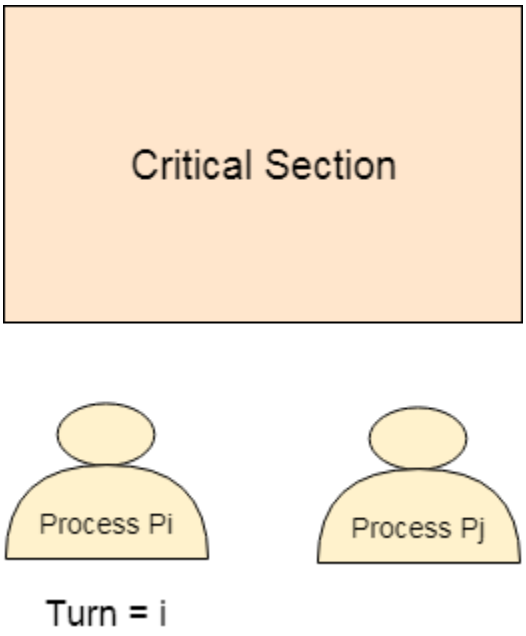
The actual problem of the lock variable approach was the fact that the process was entering in the critical section only when the lock variable is 1. More than one process could see the lock variable as 1 at the same time hence the mutual exclusion was not guaranteed there.

This problem is addressed in the turn variable approach. Now, A process can enter in the critical section only in the case when the value of the turn variable equal to the PID of the process.

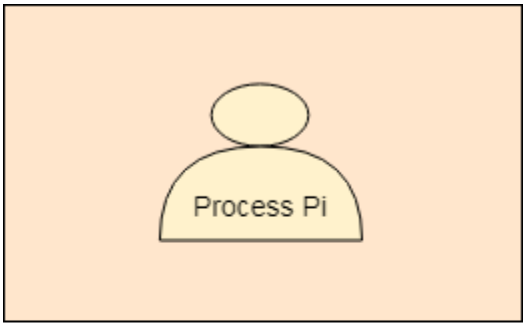
There are only two values possible for turn variable, i or j. if its value is not i then it will definitely be j or vice versa.

In the entry section, in general, the process Pi will not enter in the critical section until its value is j or the process Pj will not enter in the critical section until its value is i.

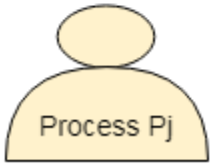
Initially, two processes Pi and Pj are available and want to execute into critical section.



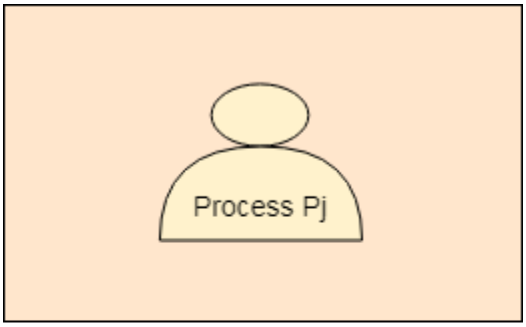
The turn variable is equal to i hence Pi will get the chance to enter into the critical section. The value of Pi remains I until Pi finishes critical section.



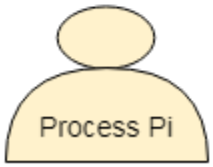
Turn = i



Pi finishes its critical section and assigns j to turn variable. Pj will get the chance to enter into the critical section. The value of turn remains j until Pj finishes its critical section.



Turn = j



Analysis of Strict Alternation approach

Let's analyze Strict Alternation approach on the basis of four requirements.

Mutual Exclusion

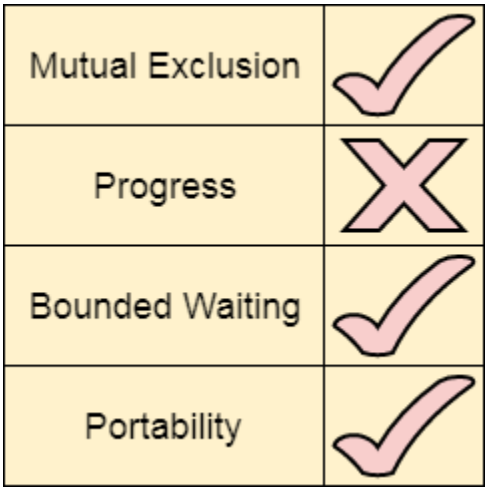
The strict alternation approach provides mutual exclusion in every case. This procedure works only for two processes. The pseudo code is different for both of the processes. The process will only enter when it sees that the turn variable is equal to its Process ID otherwise not Hence No process can enter in the critical section regardless of its turn.

Progress

Progress is not guaranteed in this mechanism. If Pi doesn't want to get enter into the critical section on its turn then Pj got blocked for infinite time. Pj has to wait for so long for its turn since the turn variable will remain 0 until Pi assigns it to j.

Portability

The solution provides portability. It is a pure software mechanism implemented at user mode and doesn't need any special instruction from the Operating System.



Paterson Solution

This is a software mechanism implemented at user mode. It is a busy waiting solution can be implemented for only two processes. It uses two variables that are turn variable and interested variable.

The Code of the solution is given below

```
1. # define N 2
2. # define TRUE 1
3. # define FALSE 0
4. int interested[N] = FALSE;
5. int turn;
6. voidEntry_Section (int process)
7. {
8.     int other;
9.     other = 1-process;
10.    interested[process] = TRUE;
11.    turn = process;
12.    while (interested [other] =True && TURN=process);
13. }
14. voidExit_Section (int process)
15. {
16.    interested [process] = FALSE;
17. }
```

Till now, each of our solution is affected by one or the other problem. However, the Peterson solution provides you all the necessary requirements such as Mutual Exclusion, Progress, Bounded Waiting and Portability.

Analysis of Peterson Solution

```
1. voidEntry_Section (int process)
2. {
3.     1. int other;
4.     2. other = 1-process;
5.     3. interested[process] = TRUE;
6.     4. turn = process;
7.     5. while (interested [other] =True && TURN=process);
8. }
9.
10. Critical Section
```

```
11.
12. voidExit_Section (int process)
13. {
14.     6. interested [process] = FALSE;
15. }
```

This is a two process solution. Let us consider two cooperative processes P1 and P2. The entry section and exit section are shown below. Initially, the value of interested variables and turn variable is 0.

Initially process P1 arrives and wants to enter into the critical section. It sets its interested variable to True (instruction line 3) and also sets turn to 1 (line number 4). Since the condition given in line number 5 is completely satisfied by P1 therefore it will enter in the critical section.

1. P1 → 1 2 3 4 5 CS

Meanwhile, Process P1 got preempted and process P2 got scheduled. P2 also wants to enter in the critical section and executes instructions 1, 2, 3 and 4 of entry section. On instruction 5, it got stuck since it doesn't satisfy the condition (value of other interested variable is still true). Therefore it gets into the busy waiting.

1. P2 → 1 2 3 4 5

P1 again got scheduled and finish the critical section by executing the instruction no. 6 (setting interested variable to false). Now if P2 checks then it are going to satisfy the condition since other process's interested variable becomes false. P2 will also get enter the critical section.

1. P1 → 6
2. P2 → 5 CS

Any of the process may enter in the critical section for multiple numbers of times. Hence the procedure occurs in the cyclic order.

Mutual Exclusion

The method provides mutual exclusion for sure. In entry section, the while condition involves the criteria for two variables therefore a process cannot enter in the critical

section until the other process is interested and the process is the last one to update turn variable.

Progress

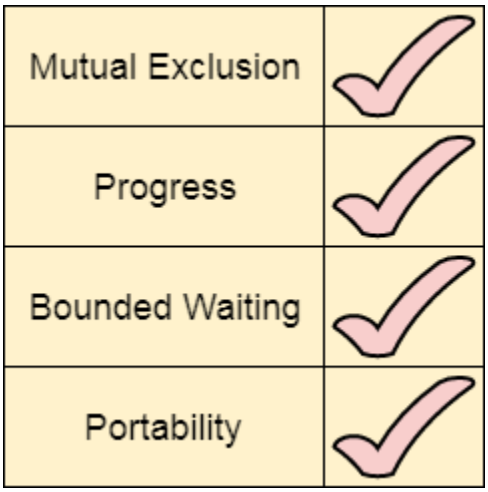
An uninterested process will never stop the other interested process from entering in the critical section. If the other process is also interested then the process will wait.

Bounded waiting

The interested variable mechanism failed because it was not providing bounded waiting. However, in Peterson solution, A deadlock can never happen because the process which first sets the turn variable will enter in the critical section for sure. Therefore, if a process is preempted after executing line number 4 of the entry section then it will definitely get into the critical section in its next chance.

Portability

This is the complete software solution and therefore it is portable on every hardware.



Synchronization Mechanism without busy waiting

All the solutions we have seen till now were intended to provide mutual exclusion with busy waiting. However, busy waiting is not the optimal allocation of resources because it

keeps CPU busy all the time in checking the while loops condition continuously although the process is waiting for the critical section to become available.

All the synchronization mechanism with busy waiting are also suffering from the priority inversion problem that is there is always a possibility of spin lock whenever there is a process with the higher priority has to wait outside the critical section since the mechanism intends to execute the lower priority process in the critical section.

However these problems need a proper solution without busy waiting and priority inversion.

Sleep and Wake

(Producer Consumer problem)

Let's examine the basic model that is sleep and wake. Assume that we have two system calls as **sleep** and **wake**. The process which calls sleep will get blocked while the process which calls will get waked up.

There is a popular example called **producer consumer problem** which is the most popular problem simulating **sleep and wake** mechanism.

The concept of sleep and wake is very simple. If the critical section is not empty then the process will go and sleep. It will be waked up by the other process which is currently executing inside the critical section so that the process can get inside the critical section.

In producer consumer problem, let us say there are two processes, one process writes something while the other process reads that. The process which is writing something is called **producer** while the process which is reading is called **consumer**.

Introduction to Semaphore in Operating Systems (OS)

Semaphore Definition

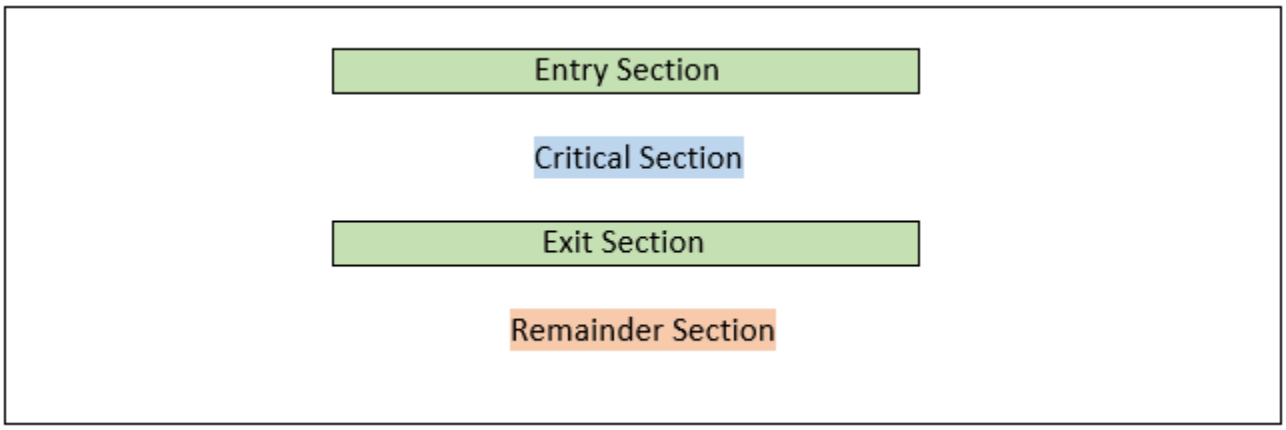
Semaphore is a Hardware Solution. This Hardware solution is written or given to critical section problem.

What is a Critical Section Problem?

The Critical Section Problem is a Code Snippet. This code snippet contains a few variables. These variables can be accessed by a few processes. There is a condition for these processes.

The condition is that only one process can only enter the critical section. Remaining Processes which are interested to enter the critical section have to wait for the process to complete its work and then enter the critical section.

Critical Section Representation



Problems in Critical Section Problems

There may be a state where one or more processes try to enter the critical state. After multiple processes enter the Critical Section, the second process try to access variable which already accessed by the first process.

Explanation

Suppose there is a variable which is also known as shared variable. Let us define that shared variable.

Here, x is the shared variable.

```
int x = 10;
```

Process 1

```
// Process 1
int s = 10;
int u = 20;
x = s + u;
```

Process 2

```
// Process 2
int s = 10;
int u = 20;
x = s - u;
```

If the process is accessed the x shared variable one after other, then we are going to be in a good position.

If Process 1 is alone executed, then the value of x is denoted as x = 30;

The shared variable x changes to 30 from 10

If Process 2 is alone executed, then the value of x is denoted as x = -10;

The shared variable x changes to -10 from 30

If both the processes occur at the same time, then the compiler would be in a confusion to choose which variable value i.e. -10 or 30. This state faced by the variable x is Data Inconsistency. These problems can also be solved by Hardware Locks

To, prevent such kind of problems can also be solved by Hardware solutions named Semaphores.

Semaphores

The Semaphore is just a normal integer. The Semaphore cannot be negative. The least value for a Semaphore is zero (0). The Maximum value of a Semaphore can be anything. The Semaphores usually have two operations. The two operations have the capability to decide the values of the semaphores.

The two Semaphore Operations are:

1. Wait ( )
2. Signal ( )

Wait Semaphore Operation

The Wait Operation is used for deciding the condition for the process to enter the critical state or wait for execution of process. Here, the wait operation has many different names. The different names are:

1. Sleep Operation
2. Down Operation
3. Decrease Operation
4. P Function (most important alias name for wait operation)

The Wait Operation works on the basis of Semaphore or Mutex Value.

Here, if the Semaphore value is greater than zero or positive then the Process can enter the Critical Section Area.

If the Semaphore value is equal to zero then the Process has to wait for the Process to exit the Critical Section Area.

This function is only present until the process enters the critical state. If the Processes enters the critical state, then the P Function or Wait Operation has no job to do.

If the Process exits the Critical Section we have to reduce the value of Semaphore

Basic Algorithm of P Function or Wait Operation

1. P (Semaphore value)
2. {
3. Allow the process to enter **if** the value of Semaphore is greater than zero or positive.
4. Do not allow the process **if** the value of Semaphore is less than zero or zero.
5. Decrement the Semaphore value **if** the Process leaves the Critical State.
6.
7. }

Signal Semaphore Operation

The Signal Semaphore Operation is used to update the value of Semaphore. The Semaphore value is updated when the new processes are ready to enter the Critical Section.

The Signal Operation is also known as:

1. Wake up Operation
2. Up Operation
3. Increase Operation
4. V Function (most important alias name for signal operation)

We know that the semaphore value is decreased by one in the wait operation when the process left the critical state. So, to counter balance the decreased number 1 we use signal operation which increments the semaphore value. This induces the critical section to receive more and more processes into it.

The most important part is that this Signal Operation or V Function is executed only when the process comes out of the critical section. The value of semaphore cannot be incremented before the exit of process from the critical section

Basic Algorithm of V Function or Signal Operation

1. V (Semaphore value)
2. {
3. If the process goes out of the critical section then add 1 to the semaphore value
4. Else keep calm until process exits
5. }

Types of Semaphores

There are two types of Semaphores.

They are:

1. Binary Semaphore

Here, there are only two values of Semaphore in Binary Semaphore Concept. The two values are 1 and 0.

If the Value of Binary Semaphore is 1, then the process has the capability to enter the critical section area. If the value of Binary Semaphore is 0 then the process does not have the capability to enter the critical section area.

2. Counting Semaphore

Here, there are two sets of values of Semaphore in Counting Semaphore Concept. The two types of values are values greater than and equal to one and other type is value equal to zero.

If the Value of Binary Semaphore is greater than or equal to 1, then the process has the capability to enter the critical section area. If the value of Binary Semaphore is 0 then the process does not have the capability to enter the critical section area.

This is the brief description about the Binary and Counting Semaphores. You will learn still more about them in next articles.

Advantages of a Semaphore

- Semaphores are machine independent since their implementation and codes are written in the microkernel's machine independent code area.
- They strictly enforce mutual exclusion and let processes enter the crucial part one at a time (only in the case of binary semaphores).
- With the use of semaphores, no resources are lost due to busy waiting since we do not need any processor time to verify that a condition is met before allowing a process access to the crucial area.
- Semaphores have the very good management of resources
- They forbid several processes from entering the crucial area. They are significantly more effective than other synchronization approaches since mutual exclusion is made possible in this way.

Disadvantages of a Semaphore

- Due to the employment of semaphores, it is possible for high priority processes to reach the vital area before low priority processes.
- Because semaphores are a little complex, it is important to design the wait and signal actions in a way that avoids deadlocks.
- Programming a semaphore is very challenging, and there is a danger that mutual exclusion won't be achieved.
- The wait ( ) and signal ( ) actions must be carried out in the appropriate order to prevent deadlocks.

Counting Semaphore

There are the scenarios in which more than one processes need to execute in critical section simultaneously. However, counting semaphore can be used when we need to have more than one process in the critical section at the same time.

A Counting Semaphore was initialized to 12. then 10P (wait) and 4V (Signal) operations were computed on this semaphore. What is the result?

1. S = 12 (initial)
2. 10 p (wait) :
3. SS = S -10 = 12 - 10 = 2
4. then 4 V :
5. SS = S + 4 =2 + 4 = 6

Hence, the final value of counting semaphore is 6.

Binary Semaphore or Mutex

In counting semaphore, Mutual exclusion was not provided because we has the set of processes which required to execute in the critical section simultaneously.

However, Binary Semaphore strictly provides mutual exclusion. Here, instead of having more than 1 slots available in the critical section, we can only have at most 1 process in the critical section. The semaphore can have only two values, 0 or 1.

What is Deadlock in Operating System (OS)?

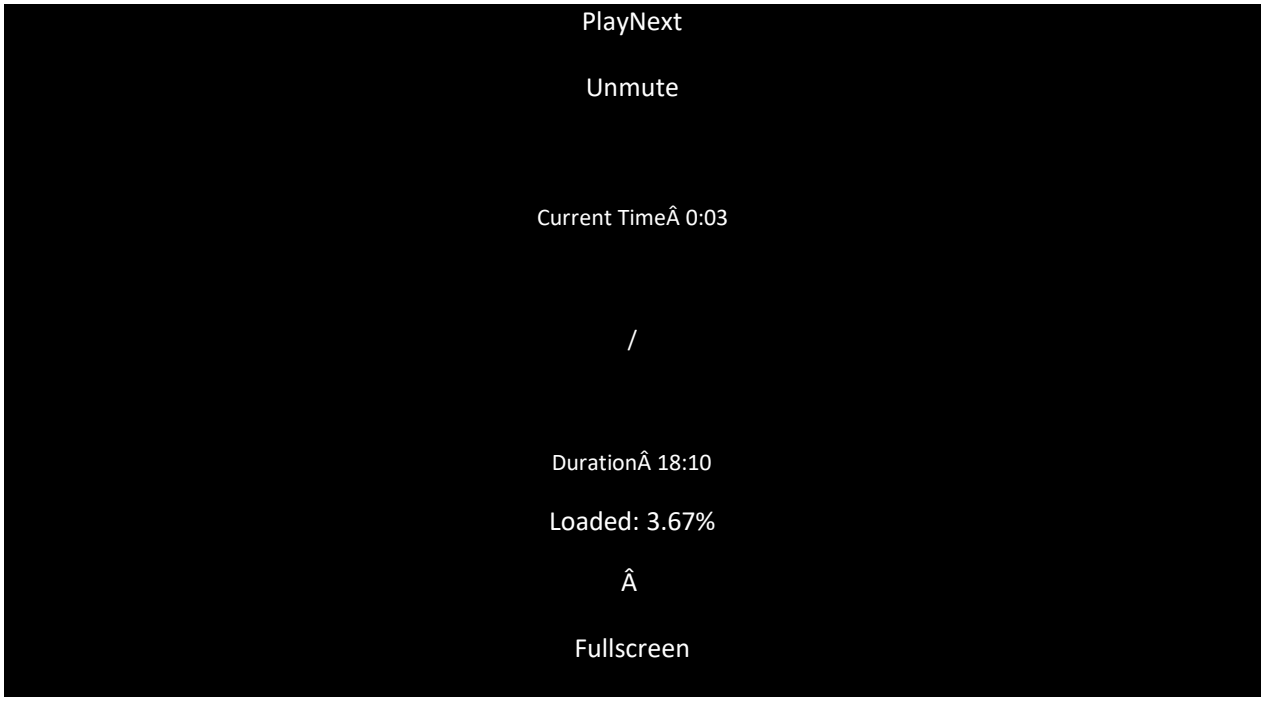
Every process needs some resources to complete its execution. However, the resource is granted in a sequential order.

1. The process requests for some resource.
2. OS grant the resource if it is available otherwise let the process waits.
3. The process uses it and release on the completion.

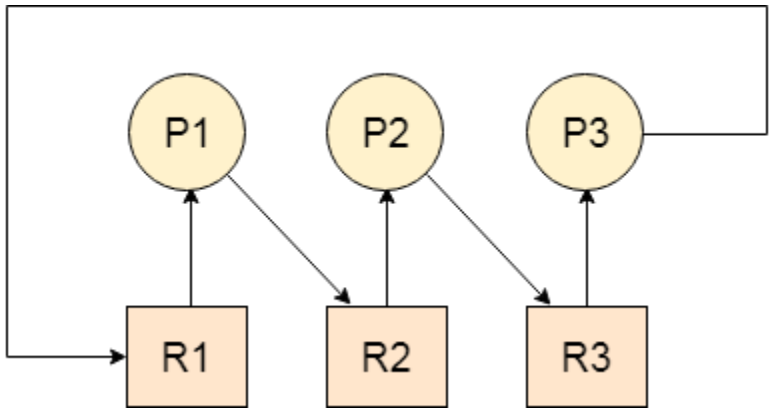
A Deadlock is a situation where each of the computer process waits for a resource which is being assigned to some another process. In this situation, none of the process gets executed since the resource it needs, is held by some other process which is also waiting for some other resource to be released.

Let us assume that there are three processes P1, P2 and P3. There are three different resources R1, R2 and R3. R1 is assigned to P1, R2 is assigned to P2 and R3 is assigned to P3.

After some time, P1 demands for R1 which is being used by P2. P1 halts its execution since it can't complete without R2. P2 also demands for R3 which is being used by P3. P2 also stops its execution because it can't continue without R3. P3 also demands for R1 which is being used by P1 therefore P3 also stops its execution.



In this scenario, a cycle is being formed among the three processes. None of the process is progressing and they are all waiting. The computer becomes unresponsive since all the processes got blocked.



Difference between Starvation and Deadlock

Sr.	Deadlock	Starvation
1	Deadlock is a situation where no process got blocked and no process proceeds	Starvation is a situation where the low priority process got blocked and the high priority processes proceed.
2	Deadlock is an infinite waiting.	Starvation is a long waiting but not infinite.
3	Every Deadlock is always a starvation.	Every starvation need not be deadlock.
4	The requested resource is blocked by the other process.	The requested resource is continuously be used by the higher priority processes.
5	Deadlock happens when Mutual exclusion, hold and wait, No preemption and circular wait occurs simultaneously.	It occurs due to the uncontrolled priority and resource management.

Necessary conditions for Deadlocks

1. **Mutual Exclusion**

A resource can only be shared in mutually exclusive manner. It implies, if two process cannot use the same resource at the same time.

2. **Hold and Wait**

   A process waits for some resources while holding another resource at the same time.

3. **No preemption**

   The process which once scheduled will be executed till the completion. No other process can be scheduled by the scheduler meanwhile.

4. **Circular Wait**

   All the processes must be waiting for the resources in a cyclic manner so that the last process is waiting for the resource which is being held by the first process.

Strategies for handling Deadlock

1. Deadlock Ignorance

Deadlock Ignorance is the most widely used approach among all the mechanism. This is being used by many operating systems mainly for end user uses. In this approach, the Operating system assumes that deadlock never occurs. It simply ignores deadlock. This approach is best suitable for a single end user system where User uses the system only for browsing and all other normal stuff.

There is always a tradeoff between Correctness and performance. The operating systems like Windows and Linux mainly focus upon performance. However, the performance of the system decreases if it uses deadlock handling mechanism all the time if deadlock happens 1 out of 100 times then it is completely unnecessary to use the deadlock handling mechanism all the time.

In these types of systems, the user has to simply restart the computer in the case of deadlock. Windows and Linux are mainly using this approach.

2. Deadlock prevention

Deadlock happens only when Mutual Exclusion, hold and wait, No preemption and circular wait holds simultaneously. If it is possible to violate one of the four conditions at any time then the deadlock can never occur in the system.

The idea behind the approach is very simple that we have to fail one of the four conditions but there can be a big argument on its physical implementation in the system.

We will discuss it later in detail.

3. Deadlock avoidance

In deadlock avoidance, the operating system checks whether the system is in safe state or in unsafe state at every step which the operating system performs. The process continues until the system is in safe state. Once the system moves to unsafe state, the OS has to backtrack one step.

In simple words, The OS reviews each allocation so that the allocation doesn't cause the deadlock in the system.

We will discuss Deadlock avoidance later in detail.

4. Deadlock detection and recovery

This approach let the processes fall in deadlock and then periodically check whether deadlock occur in the system or not. If it occurs then it applies some of the recovery methods to the system to get rid of deadlock.

We will discuss deadlock detection and recovery later in more detail since it is a matter of discussion.

Deadlock Prevention

If we simulate deadlock with a table which is standing on its four legs then we can also simulate four legs with the four conditions which when occurs simultaneously, cause the deadlock.

However, if we break one of the legs of the table then the table will fall definitely. The same happens with deadlock, if we can be able to violate one of the four necessary conditions and don't let them occur together then we can prevent the deadlock.

Let's see how we can prevent each of the conditions.

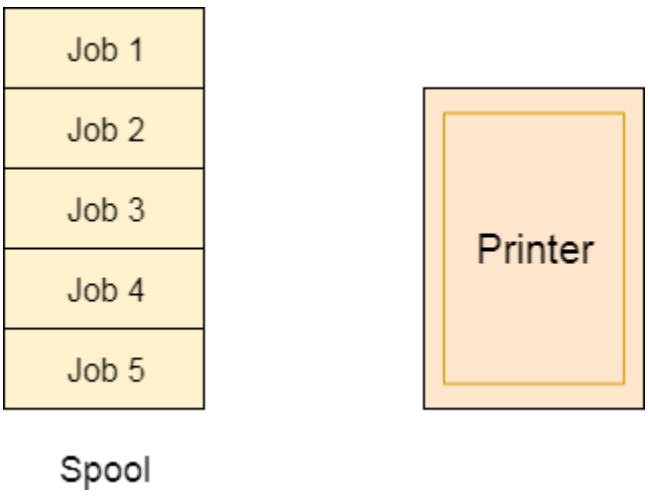
1. Mutual Exclusion

Mutual section from the resource point of view is the fact that a resource can never be used by more than one process simultaneously which is fair enough but that is the main reason behind the deadlock. If a resource could have been used by more than one process at the same time then the process would have never been waiting for any resource.

However, if we can be able to violate resources behaving in the mutually exclusive manner then the deadlock can be prevented.

Spooling

For a device like printer, spooling can work. There is a memory associated with the printer which stores jobs from each of the process into it. Later, Printer collects all the jobs and print each one of them according to FCFS. By using this mechanism, the process doesn't have to wait for the printer and it can continue whatever it was doing. Later, it collects the output when it is produced.



Although, Spooling can be an effective approach to violate mutual exclusion but it suffers from two kinds of problems.

1. This cannot be applied to every resource.
2. After some point of time, there may arise a race condition between the processes to get space in that spool.

We cannot force a resource to be used by more than one process at the same time since it will not be fair enough and some serious problems may arise in the performance. Therefore, we cannot violate mutual exclusion for a process practically.

2. Hold and Wait

Hold and wait condition lies when a process holds a resource and waiting for some other resource to complete its task. Deadlock occurs because there can be more than one process which are holding one resource and waiting for other in the cyclic order.

However, we have to find out some mechanism by which a process either doesn't hold any resource or doesn't wait. That means, a process must be assigned all the necessary resources before the execution starts. A process must not wait for any resource once the execution has been started.

!(Hold and wait) = !hold or !wait (negation of hold and wait is, either you don't hold or you don't wait)

This can be implemented practically if a process declares all the resources initially. However, this sounds very practical but can't be done in the computer system because a process can't determine necessary resources initially.

Process is the set of instructions which are executed by the CPU. Each of the instruction may demand multiple resources at the multiple times. The need cannot be fixed by the OS.

The problem with the approach is:

1. Practically not possible.
2. Possibility of getting starved will be increases due to the fact that some process may hold a resource for a very long time.

3. No Preemption

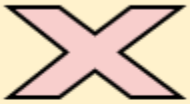
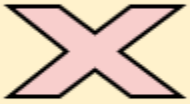
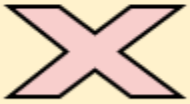
Deadlock arises due to the fact that a process can't be stopped once it starts. However, if we take the resource away from the process which is causing deadlock then we can prevent deadlock.

This is not a good approach at all since if we take a resource away which is being used by the process then all the work which it has done till now can become inconsistent.

Consider a printer is being used by any process. If we take the printer away from that process and assign it to some other process then all the data which has been printed can become inconsistent and ineffective and also the fact that the process can't start printing again from where it has left which causes performance inefficiency.

4. Circular Wait

To violate circular wait, we can assign a priority number to each of the resource. A process can't request for a lesser priority resource. This ensures that not a single process can request a resource which is being utilized by some other process and no cycle will be formed.

Condition	Approach	Is Practically Possible?
Mutual Exclusion	Spooling	
Hold and Wait	Request for all the resources initially	
No Preemption	Snatch all the resources	
Circular Wait	Assign priority to each resources and order resources numerically	

Among all the methods, violating Circular wait is the only approach that can be implemented practically.

Deadlock avoidance

In deadlock avoidance, the request for any resource will be granted if the resulting state of the system doesn't cause deadlock in the system. The state of the system will continuously be checked for safe and unsafe states.

In order to avoid deadlocks, the process must tell OS, the maximum number of resources a process can request to complete its execution.

The simplest and most useful approach states that the process should declare the maximum number of resources of each type it may ever need. The Deadlock avoidance algorithm examines the resource allocations so that there can never be a circular wait condition.

Safe and Unsafe States

The resource allocation state of a system can be defined by the instances of available and allocated resources, and the maximum instance of the resources demanded by the processes.

A state of a system recorded at some random time is shown below.

Resources Assigned

Process	Type 1	Type 2	Type 3	Type 4
A	3	0	2	2
B	0	0	1	1
C	1	1	1	0
D	2	1	4	0

Resources still needed

Process	Type 1	Type 2	Type 3	Type 4

A	1	1	0	0
B	0	1	1	2
C	1	2	1	0
D	2	1	1	2

1. E = (7 6 8 4)
2. P = (6 2 8 3)
3. A = (1 4 0 1)

Above tables and vector E, P and A describes the resource allocation state of a system. There are 4 processes and 4 types of the resources in a system. Table 1 shows the instances of each resource assigned to each process.

Table 2 shows the instances of the resources, each process still needs. Vector E is the representation of total instances of each resource in the system.

Vector P represents the instances of resources that have been assigned to processes. Vector A represents the number of resources that are not in use.

A state of the system is called safe if the system can allocate all the resources requested by all the processes without entering into deadlock.

If the system cannot fulfill the request of all processes then the state of the system is called unsafe.

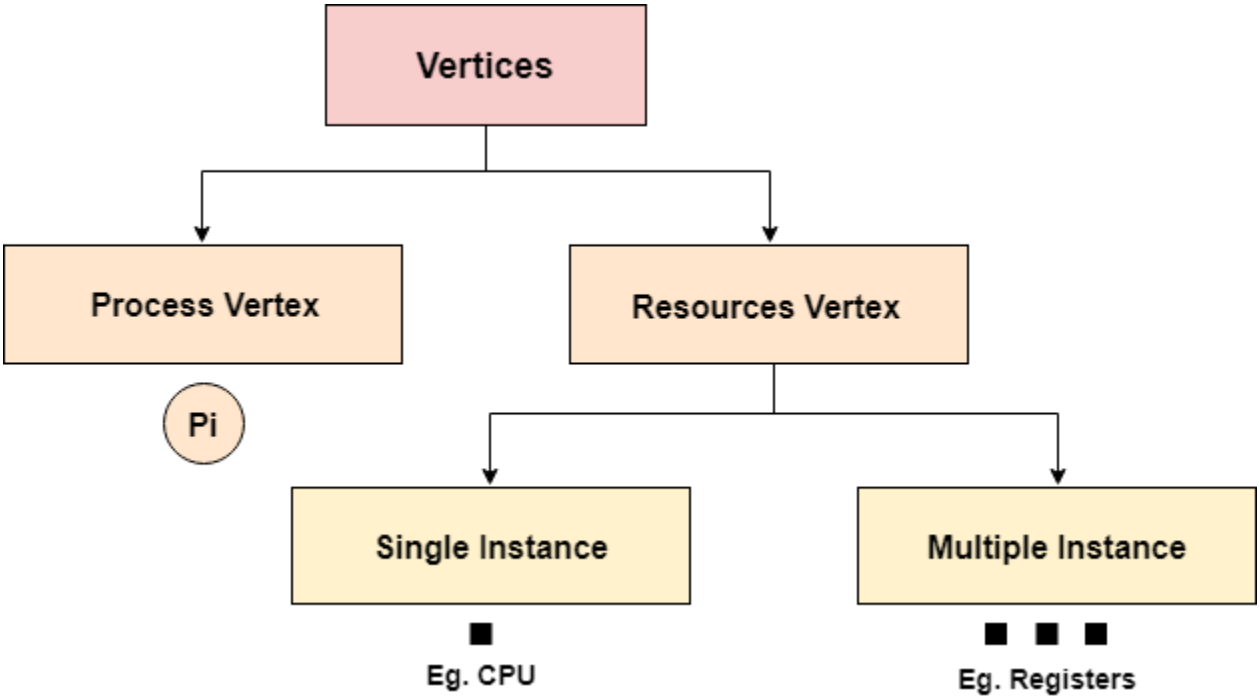
The key of Deadlock avoidance approach is when the request is made for resources then the request must only be approved in the case if the resulting state is also a safe state.

Resource Allocation Graph

The resource allocation graph is the pictorial representation of the state of a system. As its name suggests, the resource allocation graph is the complete information about all the processes which are holding some resources or waiting for some resources.

It also contains the information about all the instances of all the resources whether they are available or being used by the processes.

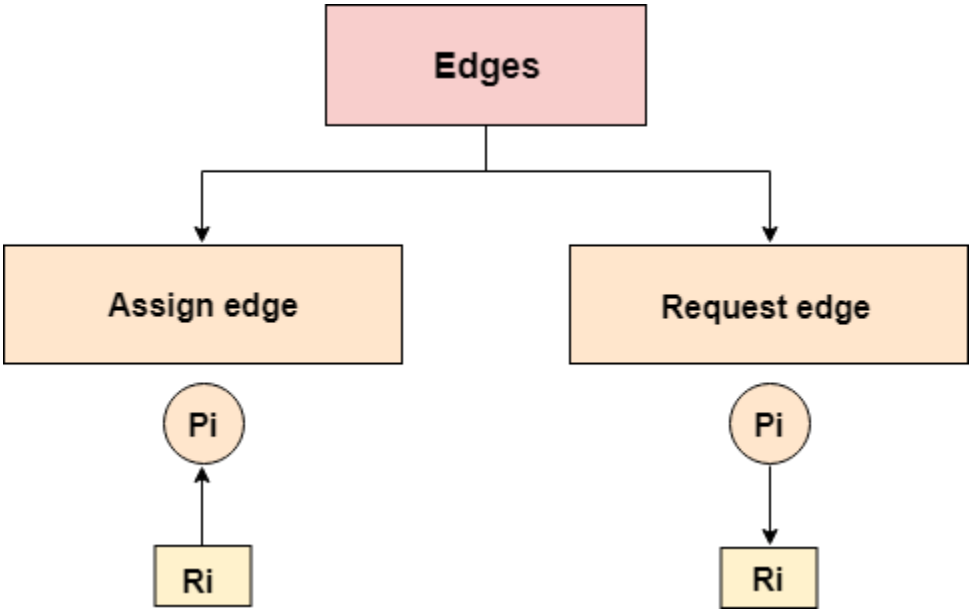
In Resource allocation graph, the process is represented by a Circle while the Resource is represented by a rectangle. Let's see the types of vertices and edges in detail.



Vertices are mainly of two types, Resource and process. Each of them will be represented by a different shape. Circle represents process while rectangle represents resource.

Backward Skip 10sPlay VideoForward Skip 10s

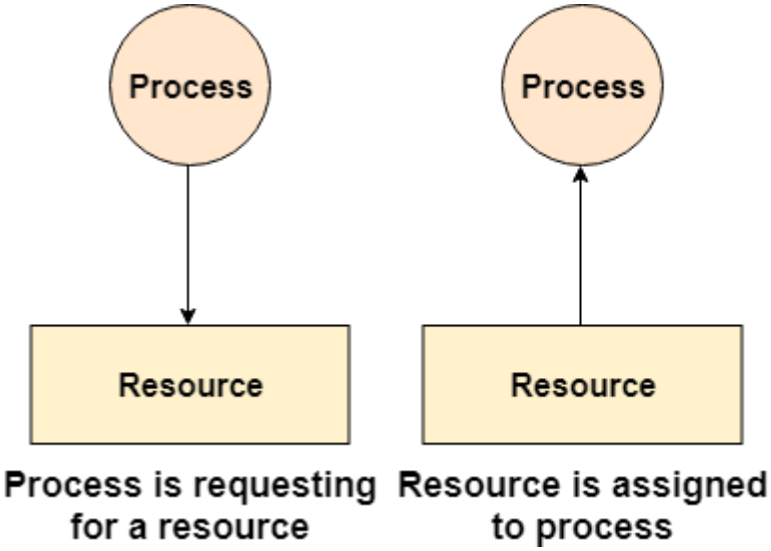
A resource can have more than one instance. Each instance will be represented by a dot inside the rectangle.



Edges in RAG are also of two types, one represents assignment and other represents the wait of a process for a resource. The above image shows each of them.

A resource is shown as assigned to a process if the tail of the arrow is attached to an instance to the resource and the head is attached to a process.

A process is shown as waiting for a resource if the tail of an arrow is attached to the process while the head is pointing towards the resource.

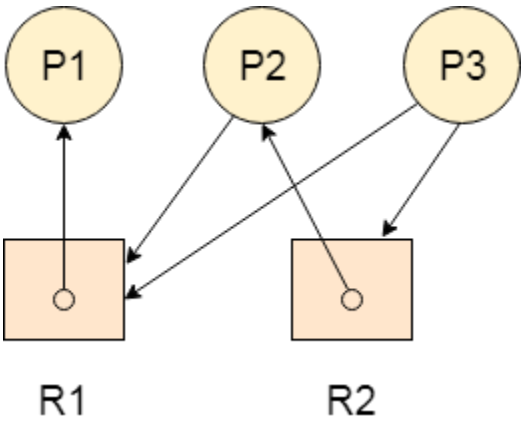


Example

Let'sconsider 3 processes P1, P2 and P3, and two types of resources R1 and R2. The resources are having 1 instance each.

According to the graph, R1 is being used by P1, P2 is holding R2 and waiting for R1, P3 is waiting for R1 as well as R2.

The graph is deadlock free since no cycle is being formed in the graph.



Banker's Algorithm in Operating System (OS)

It is a banker algorithm used to **avoid deadlock** and **allocate resources** safely to each process in the computer system. The '**S-State'** examines all possible tests or activities before deciding whether the allocation should be allowed to each process. It also helps the operating system to successfully share the resources between all the processes. The banker's algorithm is named because it checks whether a person should be sanctioned a loan amount or not to help the bank system safely simulate allocation resources. In this section, we will learn the **Banker's Algorithm** in detail. Also, we will solve problems based on the **Banker's Algorithm**. To understand the Banker's Algorithm first we will see a real word example of it.

Suppose the number of account holders in a particular bank is 'n', and the total money in a bank is 'T'. If an account holder applies for a loan; first, the bank subtracts the loan amount from full cash and then estimates the cash difference is greater than T to approve the loan amount. These steps are taken because if another person applies for a loan or withdraws some amount from the bank, it helps the bank manage and operate all things without any restriction in the functionality of the banking system.

Similarly, it works in an **operating system**. When a new process is created in a computer system, the process must provide all types of information to the operating system like upcoming processes, requests for their resources, counting them, and delays. Based on these criteria, the operating system decides which process sequence should be executed or waited so that no deadlock occurs in a system. Therefore, it is also known as **deadlock avoidance algorithm** or **deadlock detection** in the operating system.

Advantages

Following are the essential characteristics of the Banker's algorithm:

1. It contains various resources that meet the requirements of each process.
2. Each process should provide information to the operating system for upcoming resource requests, the number of resources, and how long the resources will be held.
3. It helps the operating system manage and control process requests for each type of resource in the computer system.
4. The algorithm has a Max resource attribute that represents indicates each process can hold the maximum number of resources in a system.

Disadvantages

1. It requires a fixed number of processes, and no additional processes can be started in the system while executing the process.
2. The algorithm does no longer allows the processes to exchange its maximum needs while processing its tasks.
3. Each process has to know and state their maximum resource requirement in advance for the system.
4. The number of resource requests can be granted in a finite time, but the time limit for allocating the resources is one year.

When working with a banker's algorithm, it requests to know about three things:

1. How much each process can request for each resource in the system. It is denoted by the [**MAX**] request.
2. How much each process is currently holding each resource in a system. It is denoted by the [**ALLOCATED**] resource.
3. It represents the number of each resource currently available in the system. It is denoted by the [**AVAILABLE**] resource.

Following are the important data structures terms applied in the banker's algorithm as follows:

Suppose n is the number of processes, and m is the number of each type of resource used in a computer system.

1. **Available**: It is an array of length 'm' that defines each type of resource available in the system. When Available[j] = K, means that 'K' instances of Resources type R[j] are available in the system.
2. **Max:** It is a [n x m] matrix that indicates each process P[i] can store the maximum number of resources R[j] (each type) in a system.
3. **Allocation:** It is a matrix of m x n orders that indicates the type of resources currently allocated to each process in the system. When Allocation [i, j] = K, it means that process P[i] is currently allocated K instances of Resources type R[j] in the system.

4. **Need:** It is an M x N matrix sequence representing the number of remaining resources for each process. When the Need[i] [j] = k, then process P[i] may require K more instances of resources type Rj to complete the assigned work. Nedd[i][j] = Max[i][j] - Allocation[i][j].
5. **Finish**: It is the vector of the order **m**. It includes a Boolean value (true/false) indicating whether the process has been allocated to the requested resources, and all resources have been released after finishing its task.

The Banker's Algorithm is the combination of the safety algorithm and the resource request algorithm to control the processes and avoid deadlock in a system:

Safety Algorithm

It is a safety algorithm used to check whether or not a system is in a safe state or follows the safe sequence in a banker's algorithm:

1. There are two vectors **Wok** and **Finish** of length m and n in a safety algorithm.

Initialize: Work = Available
Finish[i] = false; for I = 0, 1, 2, 3, 4... n - 1.

2. Check the availability status for each type of resources [i], such as:

Need[i] <= Work
Finish[i] == false
If the i does not exist, go to step 4.

3. Work = Work +Allocation(i) // to get new resource allocation

Finish[i] = true

Go to step 2 to check the status of resource availability for the next process.

4. If Finish[i] == true; it means that the system is safe for all processes.

Resource Request Algorithm

A resource request algorithm checks how a system will behave when a process makes each type of resource request in a system as a request matrix.

Let create a resource request array R[i] for each process P[i]. If the Resource Request$_i$ [j] equal to 'K', which means the process P[i] requires 'k' instances of Resources type R[j] in the system.

1. When the number of **requested resources** of each type is less than the **Need** resources, go to step 2 and if the condition fails, which means that the process P[i] exceeds its maximum claim for the resource. As the expression suggests:

If Request(i) <= Need
Go to step 2;

2. And when the number of requested resources of each type is less than the available resource for each process, go to step (3). As the expression suggests:

If Request(i) <= Available
Else Process P[i] must wait for the resource since it is not available for use.

3. When the requested resource is allocated to the process by changing state:

Available = Available - Request
Allocation(i) = Allocation(i) + Request (i)
$Need_i = Need_i - Request_i$

When the resource allocation state is safe, its resources are allocated to the process P(i). And if the new state is unsafe, the Process P (i) has to wait for each type of Request R(i) and restore the old resource-allocation state.

Example: Consider a system that contains five processes P1, P2, P3, P4, P5 and the three resource types A, B and C. Following are the resources types: A has 10, B has 5 and the resource type C has 7 instances.

Process	**Allocation**			**Max**			**Available**		
	A	**B**	**C**	**A**	**B**	**C**	**A**	**B**	**C**
P1	0	1	0	7	5	3	3	3	2
P2	2	0	0	3	2	2			

P3	3 0 2	9 0 2	
P4	2 1 1	2 2 2	
P5	0 0 2	4 3 3	

Answer the following questions using the banker's algorithm:

1. What is the reference of the need matrix?
2. Determine if the system is safe or not.
3. What will happen if the resource request (1, 0, 0) for process P1 can the system accept this request immediately?

Ans. 2: Context of the need matrix is as follows:

Need [i] = Max [i] - Allocation [i]
Need for P1: (7, 5, 3) - (0, 1, 0) = 7, 4, 3
Need for P2: (3, 2, 2) - (2, 0, 0) = 1, 2, 2
Need for P3: (9, 0, 2) - (3, 0, 2) = 6, 0, 0
Need for P4: (2, 2, 2) - (2, 1, 1) = 0, 1, 1
Need for P5: (4, 3, 3) - (0, 0, 2) = 4, 3, 1

Process	**Need** **A B C**
P1	7 4 3
P2	1 2 2
P3	6 0 0
P4	0 1 1

P5	4 3 1

Hence, we created the context of need matrix.

Ans. 2: Apply the Banker's Algorithm:

Available Resources of A, B and C are 3, 3, and 2.

Now we check if each type of resource request is available for each process.

Step 1: For Process P1:

Need <= Available

7, 4, 3 <= 3, 3, 2 condition is **false**.

So, we examine another process, P2.

Step 2: For Process P2:

Need <= Available

1, 2, 2 <= 3, 3, 2 condition **true**

New available = available + Allocation

(3, 3, 2) + (2, 0, 0) => 5, 3, 2

Similarly, we examine another process P3.

Step 3: For Process P3:

P3 Need <= Available

6, 0, 0 < = 5, 3, 2 condition is **false**.

Similarly, we examine another process, P4.

Step 4: For Process P4:

P4 Need <= Available

0, 1, 1 <= 5, 3, 2 condition is **true**

New Available resource = Available + Allocation

5, 3, 2 + 2, 1, 1 => 7, 4, 3

Similarly, we examine another process P5.

Step 5: For Process P5:

P5 Need <= Available

4, 3, 1 <= 7, 4, 3 condition is **true**

New available resource = Available + Allocation

7, 4, 3 + 0, 0, 2 => 7, 4, 5

Now, we again examine each type of resource request for processes P1 and P3.

Step 6: For Process P1:

P1 Need <= Available

7, 4, 3 <= 7, 4, 5 condition is **true**

New Available Resource = Available + Allocation

7, 4, 5 + 0, 1, 0 => 7, 5, 5

So, we examine another process P2.

Step 7: For Process P3:

P3 Need <= Available

6, 0, 0 <= 7, 5, 5 condition is true

New Available Resource = Available + Allocation

7, 5, 5 + 3, 0, 2 => 10, 5, 7

Hence, we execute the banker's algorithm to find the safe state and the safe sequence like P2, P4, P5, P1 and P3.

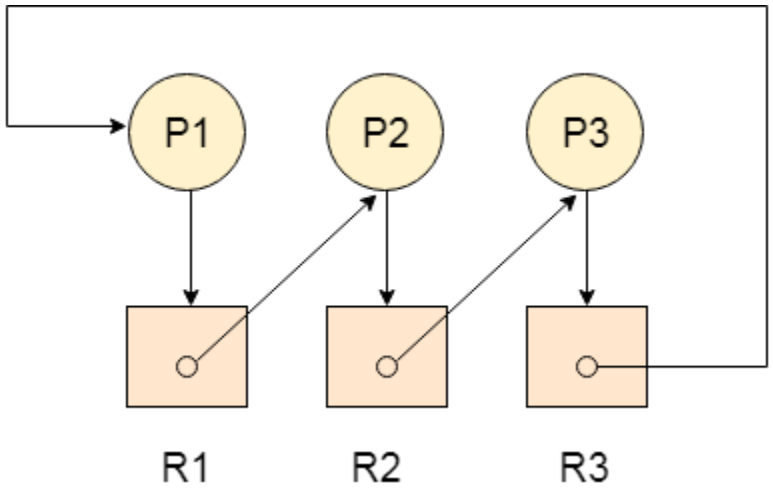
Ans. 3: For granting the Request (1, 0, 2), first we have to check that **Request <= Available**, that is (1, 0, 2) <= (3, 3, 2), since the condition is true. So the process P1 gets the request immediately.

Deadlock Detection using RAG

If a cycle is being formed in a Resource allocation graph where all the resources have the single instance then the system is deadlocked.

In Case of Resource allocation graph with multi-instanced resource types, Cycle is a necessary condition of deadlock but not the sufficient condition.

The following example contains three processes P1, P2, P3 and three resources R2, R2, R3. All the resources are having single instances each.



If we analyze the graph then we can find out that there is a cycle formed in the graph since the system is satisfying all the four conditions of deadlock.

Allocation matrix can be formed by using the Resource allocation graph of a system. In Allocation matrix, an entry will be made for each of the resource assigned. For Example,

in the following matrix, en entry is being made in front of P1 and below R3 since R3 is assigned to P1.

Process	R1	R2	R3
P1	0	0	1
P2	1	0	0
P3	0	1	0

Request Matrix

In request matrix, an entry will be made for each of the resource requested. As in the following example, P1 needs R1 therefore an entry is being made in front of P1 and below R1.

Process	R1	R2	R3
P1	1	0	0
P2	0	1	0
P3	0	0	1

Avail = (0,0,0)

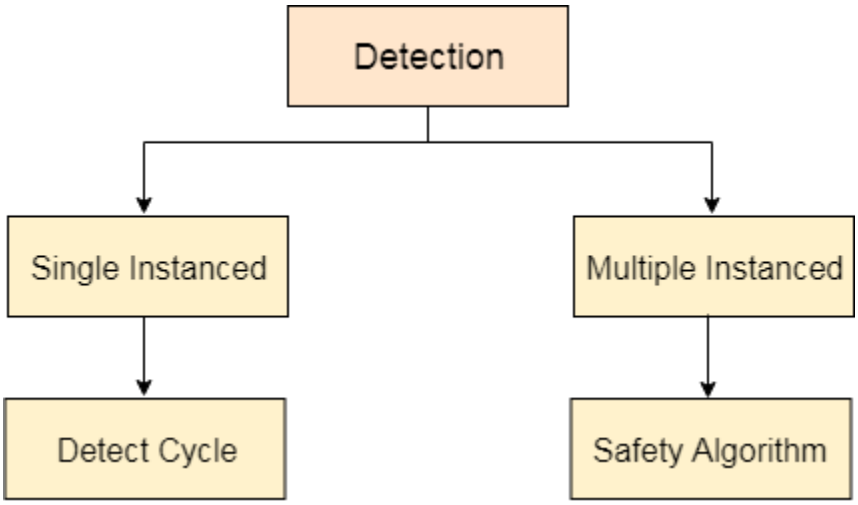
Neither we are having any resource available in the system nor a process going to release. Each of the process needs at least single resource to complete therefore they will continuously be holding each one of them.

We cannot fulfill the demand of at least one process using the available resources therefore the system is deadlocked as determined earlier when we detected a cycle in the graph.

Deadlock Detection and Recovery

In this approach, The OS doesn't apply any mechanism to avoid or prevent the deadlocks. Therefore the system considers that the deadlock will definitely occur. In order to get rid of deadlocks, The OS periodically checks the system for any deadlock. In case, it finds any of the deadlock then the OS will recover the system using some recovery techniques.

The main task of the OS is detecting the deadlocks. The OS can detect the deadlocks with the help of Resource allocation graph.



In single instanced resource types, if a cycle is being formed in the system then there will definitely be a deadlock. On the other hand, in multiple instanced resource type graph, detecting a cycle is not just enough. We have to apply the safety algorithm on the system by converting the resource allocation graph into the allocation matrix and request matrix.

In order to recover the system from deadlocks, either OS considers resources or processes.

For Resource

Preempt the resource

We can snatch one of the resources from the owner of the resource (process) and give it to the other process with the expectation that it will complete the execution and will release this resource sooner. Well, choosing a resource which will be snatched is going to be a bit difficult.

Rollback to a safe state

System passes through various states to get into the deadlock state. The operating system canrollback the system to the previous safe state. For this purpose, OS needs to implement check pointing at every state.

The moment, we get into deadlock, we will rollback all the allocations to get into the previous safe state.

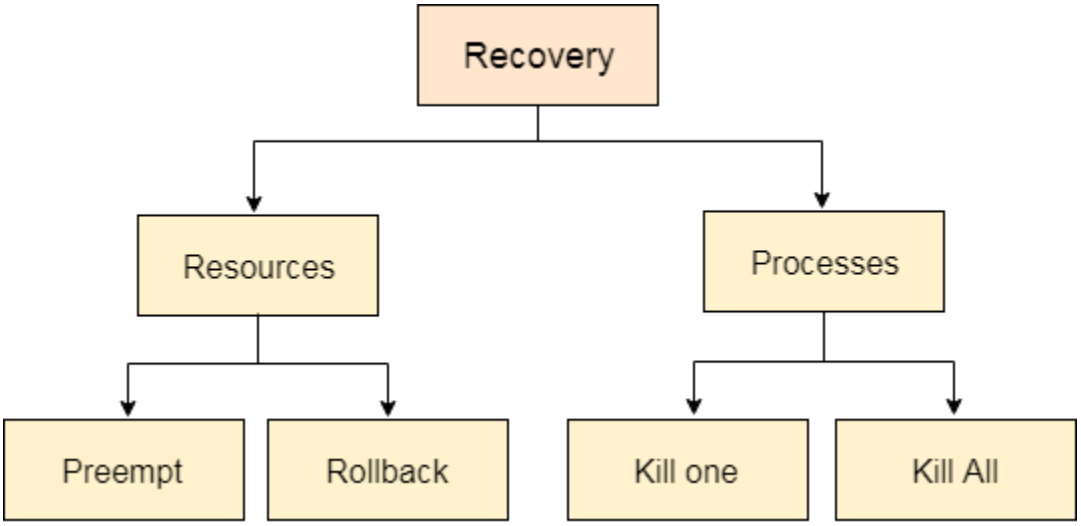
For Process

Kill a process

Killing a process can solve our problem but the bigger concern is to decide which process to kill. Generally, Operating system kills a process which has done least amount of work until now.

Kill all process

This is not a suggestible approach but can be implemented if the problem becomes very serious. Killing all process will lead to inefficiency in the system because all the processes will execute again from starting.


Recovery
Resources
Processes
Preempt
Rollback
Kill one
Kill All